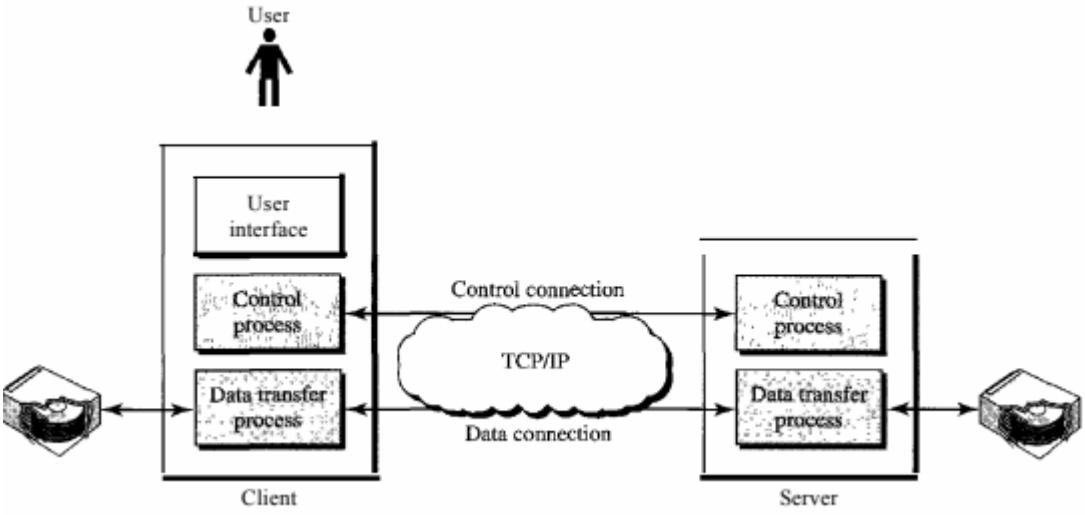
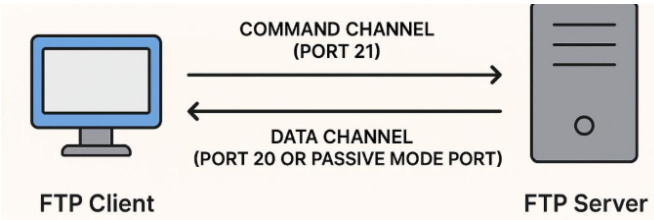
File Transfer Protocol (FTP)

FTP (File Transfer Protocol) is an application-layer protocol used to transfer files between a client and a server over a TCP/IP network. It allows users to upload, download, rename, delete and manage files on a remote computer. FTP follows the client-server architecture and uses TCP for reliable communication. FTP requires an FTP server to be running on the remote computer. Traditional FTP sends credentials and data without encryption. Use SFTP or FTPS for secure transfers.

FTP Architecture



FTP uses two separate TCP connections for communication:



1. Control Connection

- It is used to send commands and receive responses between the client and server.
- It remains active throughout the FTP session.
- It uses TCP port 21.

2. Data Connection

- It is used to transfer files between the client and server.
- It is established whenever file data needs to be transferred.
- In traditional active FTP, the server uses TCP port 20 for the data connection.

Working of FTP

1. The FTP client establishes a control connection with the FTP server using TCP port 21.

2. The user provides login credentials (username and password).
3. After successful authentication, the server allows the client to access files.
4. The client sends commands to the server to upload, download or manage files.
5. A separate data connection is established to transfer the requested files.
6. After file transfer is completed, the data connection is closed.
7. The control connection remains active until the user terminates the FTP session.

FTP supports two modes of establishing data connections:

- Active Mode: The server initiates the data connection to the client.
- Passive Mode: The client initiates the data connection to the server.

Common FTP commands

Command	Purpose
`open`	Connect to FTP server
`user`	Enter username
`dir`	List files
`get`	Download a file
`put`	Upload a file
`delete`	Delete a file
`bye`	Close FTP connection

Some scenario where FTP is used

Users / Organizations	How they use FTP
Web hosting companies	Upload website files, images and web pages to hosting servers.
Banks and financial institutions	Exchange reports and transaction files between systems, often using SFTP.
Hospitals	Transfer reports, scanned documents and data between systems using secure file transfer solutions.
Software companies	Transfer software packages, application files and backups.
Media and broadcasting companies	Transfer large video, audio and graphics files.
Manufacturing industries	Exchange inventory, order and supply-chain files with business partners.
Network administrators	Transfer configuration files, logs and backups between servers.